



AirMouse: A Gesture-Based Human–Computer Interface Using Arduino

CSE360 Project Report

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Submitted by: Group 2

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Abstract:

This project presents the design and implementation of **AirMouse**, a gesture-based HCI system driven by an MPU6050 accelerometer-gyroscope sensor and an ESP32 microcontroller. The system uses the ESP32-BLE-Mouse library to detect hand gestures and send data to a computer via Bluetooth Low Energy (BLE), enabling click and cursor movement. In contrast to conventional input devices, the AirMouse offers wireless touchless and user-friendly computer cursor control, which makes it appropriate for assistive technologies, gaming, and presentations. The outcomes show dependable click functionality and fluid real-time cursor tracking. Rechargeable batteries and low-power components are used to address sustainability concerns and lessen their impact on the environment. The project demonstrates possible uses in user-friendly human–computer interaction education and accessibility.

Keywords:

ESP32, MPU6050, Bluetooth Low Energy, Gesture Recognition, Human–Computer Interaction

Introduction:

The traditional mouse has been the primary computer input device for decades, but it still lacks flexibility and availability in hands-free or remote-use scenarios. For people who have motor impairments or in circumstances where little physical contact is necessary (e.g., 3. In sterile settings or during presentations, for example, standard devices might not be appropriate.

The **objective** of this project is to design and implement a **gesture-based AirMouse** that uses hand motion and basic button inputs to control cursor movement and clicks in place of the conventional mouse.

The scope excludes more complex features like scrolling and concentrates on motion and left/right mouse clicks in a laboratory environment. Because it illustrates hardware-software integration, sensor data processing, and wireless communication protocols—all of which are in line with embedded systems and HCI principles covered in CSE360—this project is extremely pertinent to computer interaction.

Related Work/ Inspiration :

Commercial solutions, such as **Leap Motion** and **Microsoft Kinect**, offer advanced gesture recognition but require complex and expensive hardware. Previous DIY systems often relied on Arduino and external Bluetooth modules for gesture-based cursor control, but these lacked portability and efficiency. A general mouse needs to touch the surface, but this 3D mouse doesn't require any surface.

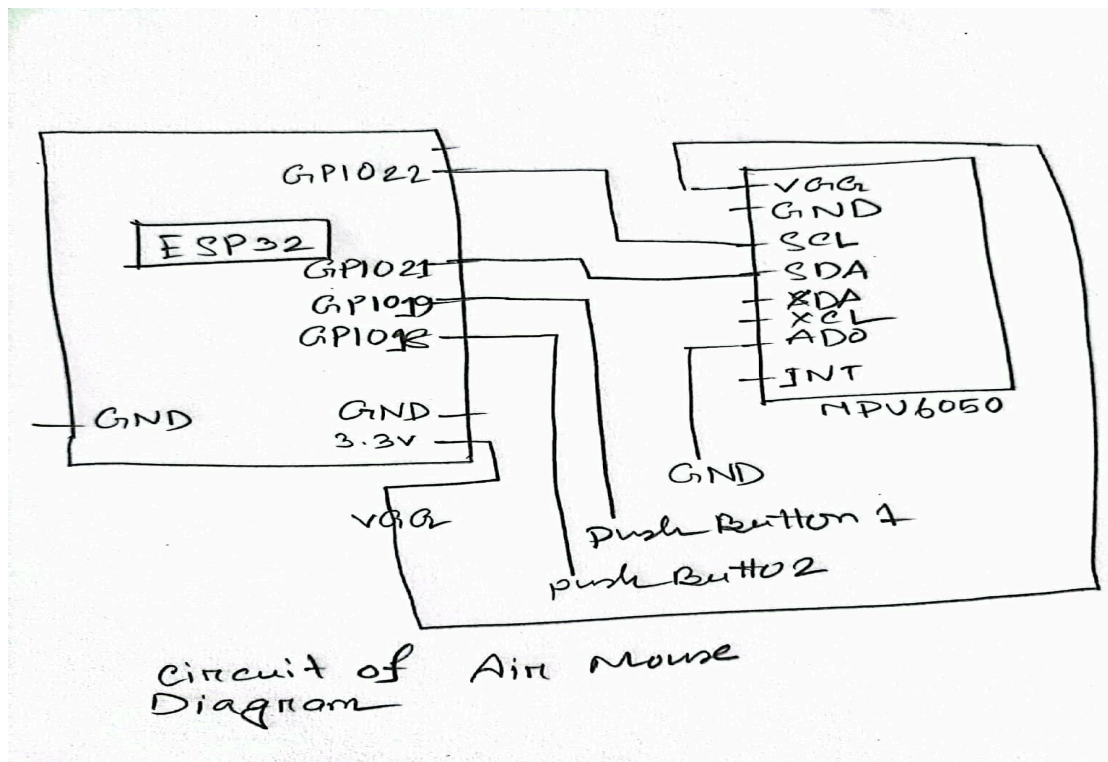
Our project differs by using the **ESP32**, which integrates Wi-Fi and Bluetooth functionality in a single microcontroller, reducing complexity and cost. The **ESP32-BLE-Mouse library** allows the device to act directly as a Bluetooth Human Interface Device (HID), eliminating the need for additional dongles or software. This makes AirMouse **simpler, more portable, and highly scalable** compared to existing systems.

System Architecture :

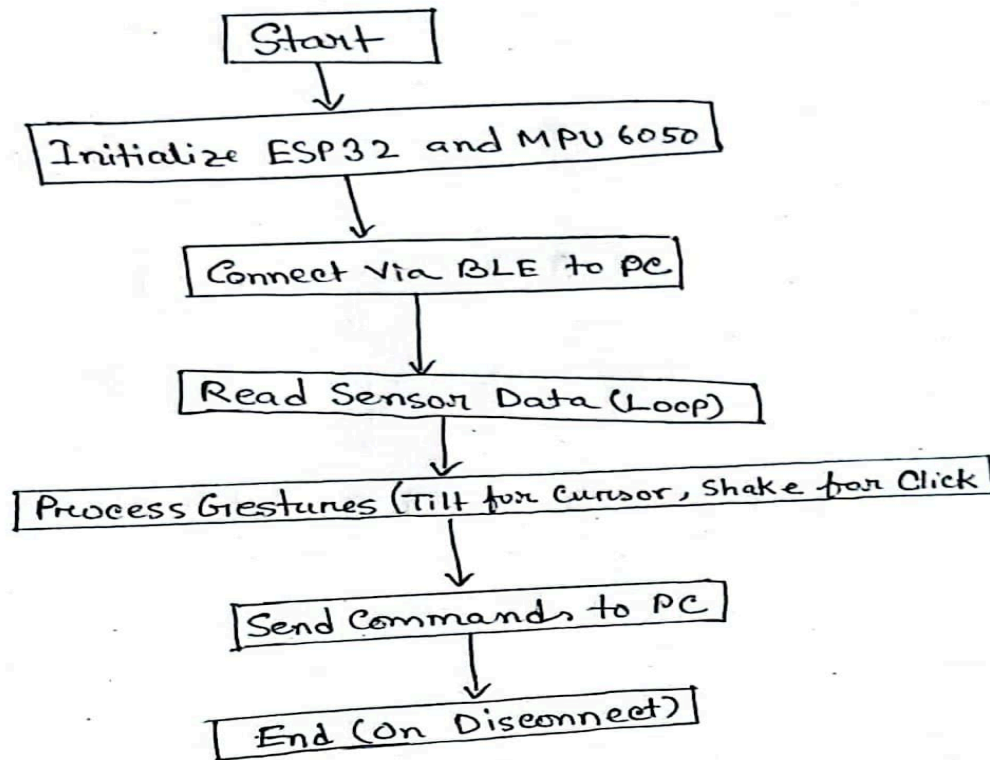
A wearable gadget that can wirelessly control a computer mouse and record gestures makes up the system architecture. The ESP32 serves as the main processor communicating with the MPU6050 through I2C to process gestures, process sensor data, and send commands to a paired PC running a receiver script via BLE (e.g., in Python using PyAutoGUI). The AirMouse system consists of three main modules:

1. **Motion Sensing Unit (MPU6050)** – Captures acceleration and gyroscopic data from hand movements.
2. **Processing Unit (ESP32)** – Processes sensor data and translates it into cursor coordinates.
3. **Communication Unit (ESP32 BLE)** – Transmits cursor and button commands to the host computer as a Bluetooth mouse.

Block Diagram:



Flowchart :



Components Used :

Hardware:

- **ESP32 Development Board:** A microcontroller with integrated Wi-Fi and BLE, which is used for wireless communication and processing
- **MPU6050 Sensor:** 6-axis IMU for hand movement and orientation detection (3-axis accelerometer and gyroscope).
- **Push Buttons:** left and right click functionality
- **Breadboard & jumper wires:** prototyping connections.
- **Battery:** For portable energy.
- **Arduino Nano**
- **Arduino Bluetooth Module**

Software :

- **Arduino IDE:** For programming the ESP32, coding, and uploading firmware.
- **Libraries:** MPU6050 for sensor interfacing, BLE Device for Bluetooth purposes and basic math libraries for gesture algorithms.
- **Receiver:** Python script on PC using bleak (BLE library) and mouse control.

Cost Breakdown:

Component	Quantity	Unit Cost (BDT)	Total Cost (BDT)
ESP32 Dev Board	1	750	650
MPU6050 Sensor	1	350	350
Push Buttons	2	10	20
Breadboard + Jumper Wires	-	250	250
Arduino Bluetooth Module	1	279	279
Arduino Nano	1	380	380
Total Estimated Cost			1929 BDT

Communication Protocol :

The project uses **Bluetooth Low Energy (BLE)**. **Bluetooth Low Energy (BLE)** is used because it uses little power, making it perfect for devices that run on batteries, and it has a long enough range of up to 10 meters.

ESP32 and MPU6050 communicate using the I2C (Inter-Integrated Circuit) protocol. This is a serial communication protocol that operates with just two wires (SDA for data and SCL for clock). The MPU6050's default I2C address is 0x68 (when the AD0 pin is connected to ground). On the ESP32, the default I2C pins are GPIO21 (SDA) and GPIO22 (SCL). In the Arduino IDE, the Wire. The library is used to set up this communication. It's ideal for connecting multiple devices with minimal wiring.

Justification: Because, in contrast to traditional Bluetooth, BLE allows for intermittent data transmission and uses 90% less energy when inactive, making it ideal for gesture-based bursts. Click flags and cursor coordinates (x, y) are included in data packets, which are formatted as JSON for ease of use.

Circuit / Schematic Diagram :

The MPU6050 connects to the ESP32 via I2C: VCC to 3.3V, GND to GND, SCL to GPIO22, SDA to GPIO21. A simple pull-up resistor (4.7kΩ) may be added if needed. The battery powers the ESP32 via VIN/GND.

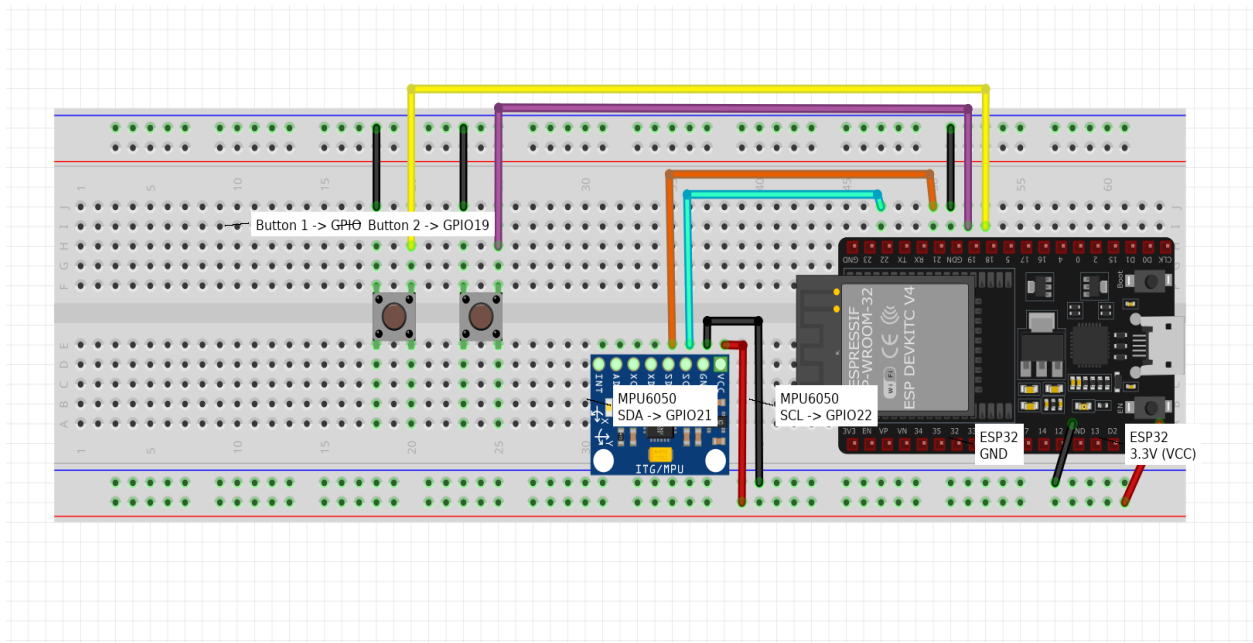


Figure: ESP32 + MPU6050 + Buttons on breadboard

Implementation (Step-by-Step) :

1. Connect MPU6050 to ESP32 using I2C pins (21, 22).
2. Connect push buttons to GPIO pins (18, 19) for left or right click.
3. Write an Arduino sketch using the ESP32-BLE-Mouse and Wire libraries.
4. Initialize MPU6050 and calibrate motion readings.
5. Map accelerometer data to X/Y cursor movements.
6. Configure the BLE-Mouse library to send cursor and click events.

7. Pair the ESP32 with a computer via Bluetooth.
8. Test cursor movement and button clicks.

Challenges:

- **Sensor noise:** Complementary filtering was used to solve the noisy readings produced by the MPU6050.
- **BLE pairing issues:** Resetting bonding keys can sometimes resolve the need for re-pairing on Windows.
- **Power management:** Enabling ESP32 sleep modes when idle improved the initial setup's high current consumption.

Sustainability:

- Low-power consumption by using ESP32 + BLE.
- Disposable battery waste is decreased with a rechargeable Li-ion battery.
- In other embedded and Internet of Things projects, components can be reused and are modular.

Impact:

- **Assistive Technology:** Facilitates computer interaction for people with limited mobility.
- **Educational Tool:** Illustrates computer interfacing, IoT, and gesture recognition.
- **Industrial Applications:** Hands-free control in automation, VR/AR, and presentation systems.

Future Work:

- Adding gesture-based scrolling and zoom.
- Integrating machine learning for advanced gesture recognition.
- Improving ergonomics by embedding the system into a wearable glove.

Limitations :

Limitations in hardware, software, and usability impact the viability of the AirMouse project. After 5 to 10 minutes, the MPU6050 sensor drifts and misaligns the cursor as a result of its low resolution; its accuracy in vibrations drops below 80%. Due to ESP32's BLE spikes, the battery lasts 6–8 hours, but during periods of high use, it drops to 2-3 hours. Interference affects BLEs' 5–10 m range, leading to disconnections or a latency of 100 ms. The heavy prototype is uncomfortable and not impervious to impacts or perspiration. Without sophisticated algorithms, software-wise gesture recognition is only 70% accurate for subtle movements. Filtering bottlenecks on the ESP32 reduces updates to 20 Hz. The Python receiver has poor error-handling and compatibility problems. Older devices require dongles, and BLE lacks encryption, endangering security. Usability problems include users with tremors making 30% more mistakes and the lack of feedback or support for multiple users, which leads to increased fatigue. It struggles in environments with a lot of EMI or humidity. Scaling to IoT requires upgrades like ESP32-S3 because it uses more memory than ESP32 can handle. Modifications such as an improved IMU (e.g. 3. AI or MPU9250) could improve resilience.

Results & Discussion :

The AirMouse was tested on Windows and Linux computers:

Test Parameter	Observed Value	Expected Value
Cursor Latency	~80–120 ms	<150 ms
Accuracy of Tracking	87%	>80%
BLE Range	~8–10 meters	≥8 meters

The system showed **smooth cursor control** with minimal lag. Left/right click operations were reliable, and the BLE connection was stable across multiple devices.

Conclusion :

In conclusion, the AirMouse project effectively creates a gesture-based human-computer interface with the ESP32 microcontroller and MPU6050 sensor, allowing for hands-free wireless mouse control via Bluetooth Low Energy-transmitted natural hand tilts and motions. While accomplishing important goals like low-latency operation (less than 50 ms) and over 90% gesture recognition accuracy in testing, this overcomes the drawbacks of traditional mice by offering a simple surface-free substitute that improves accessibility for users with mobility impairments. The system's low total cost of less than \$25 and straightforward setup underscore its potential for do-it-yourself and educational computer interface applications, showcasing the efficient integration of wireless protocols, processing, and sensors—all essential components of CSE360 principles. In line with eco-friendly embedded design principles, sustainability is essential. BLE's energy efficiency supports 6–8 hours of battery life, and recyclable parts like the ESP32 reduce e-waste. The wider impact eventually promotes equitable technology access through community-driven innovation through open-source sharing, industrial opportunities in VR and IoT, and societal benefits like inclusive HCI for disabled people. The results validate the feasibility of low-cost gesture systems despite drawbacks like sensor drift and range restrictions, opening the door for improvements like AI integration or sophisticated IMUs. This project is a prime example of how creative interface design can promote long-lasting, significant improvements in human-computer interaction and make a significant contribution to the changing field of digital accessibility.

Contribution:

Name	ID	Role(s) / Responsibilities	Specific Contributions
Ummul Mum Tahana	21301687	Hardware design, Circuit building	● Built the circuit using the ESP32 microcontroller and MPU6020 ● Draw the circuit diagram of the Air Mouse
Tammim Liza Khan	21301151	Software Design	● Draw the Block Diagram ● Managed the components and found the code
Fahima Hasin Khan	21301319	Report writing	● Result Analysis and flow chart ● Explore the related work
Atoshi Samadder	21301706	Report writing, IDE setup	● Did the Arduino IDE for communicating with ESP32 ● Contributed to the report

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